

Furniture CRM

Deployment & Operations Guide

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01 — System Requirements

Software

Node.js	16 or newer (v18 LTS recommended). Check: <code>node --version</code>
npm	Comes with Node.js. Check: <code>npm --version</code>
OS	Linux (Ubuntu 22.04 recommended), macOS 12+, or Windows 10+ with WSL2
RAM	Minimum 512 MB free. 1 GB+ recommended for production
Disk	At least 200 MB free for the app, logs, and database backups
Port	3456 (default). Must be open in firewall for external access

Optional (for production)

Nginx	Reverse proxy + SSL termination. Install: <code>sudo apt install nginx</code>
Certbot	Free SSL certificates from Let's Encrypt. Install: <code>sudo apt install certbot</code>
PM2	Process manager — keeps the app running after crashes and reboots
UFW Firewall	<code>sudo apt install ufw</code> — block unused ports

02 — Installation — Step by Step

Note

These steps assume a fresh Ubuntu 22.04 VPS. Run all commands as a non-root user with sudo access.

Install Node.js 18 on Ubuntu

```
# Add NodeSource repository for Node 18

curl -fsSL https://deb.nodesource.com/setup_18.x | sudo -E bash -

# Install Node.js

sudo apt-get install -y nodejs

# Verify installation

node --version # should print v18.x.x

npm --version # should print 9.x.x
```

Upload the CRM folder to your server

```
# From your local Mac – upload the CRM folder to the server

scp -r ~/Desktop/CRM user@YOUR_SERVER_IP:/home/user/crm

# Or use rsync for large transfers

rsync -avz --exclude node_modules --exclude data/db.json \
~/Desktop/CRM/ user@YOUR_SERVER_IP:/home/user/crm/
```

Install dependencies on the server

```
# SSH into your server

ssh user@YOUR_SERVER_IP

# Go to the CRM folder
```

```
cd /home/user/crm

# Install Node.js packages

npm install

# Confirm no critical errors

npm audit --audit-level=critical
```

Create the data directory

```
# The database lives in ./data/ (not served by web)

mkdir -p data

chmod 700 data # only your user can read it
```

03 — Environment Configuration (.env)

Security

The .env file contains your secret keys. Never commit it to Git, never share it, and set strict permissions on the server.

Create the .env file

```
# On your server, inside /home/user/crm/
```

```
nano .env
```

Contents of .env

```
# Server

PORT=3456

NODE_ENV=production

# JWT Secret – generate a new one with the command below

JWT_SECRET=paste_your_generated_secret_here

# Optional: your domain (restricts CORS to your site only)

ALLOWED_ORIGIN=https://yourdomain.com
```

Generate a strong JWT secret

```
# Run this on your server to generate a secure random secret

node -e "console.log(require('crypto').randomBytes(48).toString('hex'))"

# Copy the output and paste it as JWT_SECRET in .env
```

Protect the .env file

```
# Only your user can read it – no one else

chmod 600 .env
```

```
# Verify permissions (should show -rw-----)
```

```
ls -la .env
```

04 — Running the Server

Quick test (development only)

[illegible]

Production — PM2 (recommended)

```
# Install PM2 globally

sudo npm install -g pm2


# Start the CRM using the included PM2 config

pm2 start ecosystem.config.js --env production


# Check status

pm2 status


# View live logs

pm2 logs crm


# Restart after code changes

pm2 reload crm
```

Auto-start on server reboot

```
# Generate the startup script (run as your user, not root)

pm2 startup

# PM2 will print a command – copy it and run it with sudo

# It looks like: sudo env PATH=... pm2 startup systemd -u user

# Save the current process list

pm2 save

# Test it works after reboot

sudo reboot

# After reconnecting: pm2 status → should show "online"
```

PM2 useful commands

pm2 status	See all running processes and their CPU/memory
pm2 logs crm	Live log stream (Ctrl+C to exit)
pm2 logs crm --lines 100	Last 100 log lines
pm2 reload crm	Zero-downtime reload after code update
pm2 stop crm	Stop the server
pm2 delete crm	Remove from PM2 list
pm2 monit	Real-time dashboard with CPU, memory, logs

05 — Deploying to a Live VPS

Recommended VPS providers

DigitalOcean	droplets.digitalocean.com — \$6/mo Droplet (1GB RAM) is enough
Hetzner	hetzner.com — cheapest in Europe, €4/mo CX11
Vultr	vultr.com — \$6/mo, many locations worldwide
Linode / Akamai	linode.com — \$5/mo Nanode

Firewall setup (UFW)

```
sudo ufw allow OpenSSH # keep SSH open!

sudo ufw allow 80/tcp # HTTP (for Nginx + Let's Encrypt)

sudo ufw allow 443/tcp # HTTPS

sudo ufw deny 3456/tcp # block direct Node.js access – use Nginx instead

sudo ufw enable

# Check rules

sudo ufw status verbose
```

Important

Never expose port 3456 directly to the internet. Always put Nginx in front. Nginx handles HTTPS and forwards traffic to Node.js internally.

Point your domain to the server

In your domain registrar's DNS settings, create an **A record**:

Type	A
Name	@ (or crm.yourdomain.com for a subdomain)
Value	YOUR_SERVER_IP_ADDRESS
TTL	3600 (or Auto)

06 — HTTPS / SSL Setup with Nginx

Install Nginx and Certbot

```
sudo apt update

sudo apt install -y nginx certbot python3-certbot-nginx
```

Create Nginx config for the CRM

```
sudo nano /etc/nginx/sites-available/crm
```

Paste this into the file:

```
server {

    listen 80;

    server_name yourdomain.com www.yourdomain.com;

    # Security headers

    add_header X-Frame-Options "SAMEORIGIN" always;

    add_header X-Content-Type-Options "nosniff" always;

    add_header Referrer-Policy "strict-origin-when-cross-origin" always;

    # Gzip

    gzip on;

    gzip_types text/plain text/css application/json application/javascript;

    # Proxy to Node.js

    location / {

        proxy_pass http://127.0.0.1:3456;

        proxy_http_version 1.1;

        proxy_set_header Upgrade $http_upgrade;
```

```
proxy_set_header Connection 'upgrade';

proxy_set_header Host $host;

proxy_set_header X-Real-IP $remote_addr;

proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;

proxy_set_header X-Forwarded-Proto $scheme;

proxy_read_timeout 30s;

proxy_send_timeout 30s;

client_max_body_size 12M;

}

}
```

Enable the site and get SSL certificate

```
# Enable the site

sudo ln -s /etc/nginx/sites-available/crm /etc/nginx/sites-enabled/

# Test config syntax

sudo nginx -t

# Reload Nginx

sudo systemctl reload nginx

# Get free SSL certificate from Let's Encrypt

sudo certbot --nginx -d yourdomain.com -d www.yourdomain.com

# Certbot automatically:

# 1. Gets the certificate

# 2. Updates your Nginx config for HTTPS
```

```
# 3. Sets up auto-renewal (runs every 90 days)
```

```
# Verify auto-renewal works
```

```
sudo certbot renew --dry-run
```

Result

Your CRM will be available at <https://yourdomain.com> with a valid SSL certificate. HTTP requests automatically redirect to HTTPS.

07 — Security Overview

What is already protected in your CRM

Severity	Threat	Protection Applied
CRITICAL	Database file (db.json) exposed to web	Moved to data/ folder. All .json/.env requests return 403 Forbidden
CRITICAL	JWT secret hardcoded in source code	Moved to .env file. Server refuses to start without a strong secret
HIGH	Brute-force login attacks	5 failed logins locks the account for 15 minutes
HIGH	Rate limiting / DDoS on API	Auth: 20 req/15min · API: 120 req/min · SMTP: 5 req/min
HIGH	Missing security headers (XSS, Clickjacking)	Helmet.js adds X-Frame-Options, CSP, X-XSS-Protection, HSTS
MEDIUM	Server error details leaked to client	Production mode returns generic messages. Details only in server logs
MEDIUM	SMTP password returned in API response	GET /api/smtp strips the password field before sending to browser
MEDIUM	No input validation — oversized payloads	All inputs: trimmed, length-capped, type-validated. API: 500KB limit
MEDIUM	Timing attacks on login (user enumeration)	bcrypt runs even for unknown emails to prevent timing-based detection
LOW	Passwords stored in plaintext	bcrypt with 12 rounds (slow hash, resistant to rainbow tables)
LOW	Sensitive files indexed by search engines	robots meta tag: noindex, nofollow

Additional hardening recommendations

SSH keys only	Disable password SSH login: PasswordAuthentication no in /etc/ssh/sshd_config
Fail2ban	sudo apt install fail2ban — auto-bans IPs with repeated SSH failures
Regular updates	sudo apt update && sudo apt upgrade — run monthly at minimum
2FA for server SSH	Use Google Authenticator PAM module for SSH two-factor auth

Change SSH port

Edit /etc/ssh/sshd_config: Port 2222 — reduces automated attacks

08 — Database & Backups

Database location

File	/home/user/crm/data/db.json
Format	JSON (lowdb flat-file database)
Contents	Users, CRM companies, tasks, notes, activity log, SMTP settings
Access	chmod 700 data/ — only your server user can read it
Web access	Blocked — returns 403 Forbidden if requested via browser

Manual backup

```
cd /home/user/crm

./backup.sh

# Output:

# ■ Backup saved: backups/db_2026-06-09_03-00.json

# ■ Old backups pruned (keeping 30)
```

Automatic daily backup with cron

```
# Open cron editor

crontab -e

# Add this line to run backup every day at 3:00 AM

0 3 * * * /home/user/crm/backup.sh >> /home/user/crm/logs/backup.log 2>&1
```

Restore from backup

```
# Stop the server first

pm2 stop crm
```

```
# Replace database with backup

cp backups/db_2026-06-09_03-00.json data/db.json


# Start server again

pm2 start crm


# Verify it loaded correctly

curl http://localhost:3456/api/health
```

Off-site backup (optional but strongly recommended)

```
# Copy latest backup to another server or S3 bucket

scp backups/db_$(date +%Y-%m-%d)*.json backup@BACKUP_SERVER:/backups/crm/


# Or use rclone to sync to Google Drive / Dropbox / S3

# Install: sudo apt install rclone

# Config: rclone config

rclone copy /home/user/crm/backups remote:crm-backups
```

09 — Email (SMTP) Setup

Email is used to notify team members when a task is assigned to them. Configure it inside the CRM under Tasks → Email Settings.

Gmail (recommended)

1 Enable 2-Step Verification

Go to myaccount.google.com/security → 2-Step Verification → Turn On

2 Create an App Password

Go to myaccount.google.com/apppasswords → Select app: Mail → Generate. Copy the 16-character code.

3 Enter settings in CRM

Tasks → Email Settings: • Host: `smtp.gmail.com` • Port: 587 • Email: `yourname@gmail.com` • Password: paste the App Password (NOT your Gmail password)

4 Click Save & Test

The server sends a test connection. If it shows ■ Connected, email is working.

Other email providers

Outlook / Hotmail

Host: `smtp-mail.outlook.com` Port: 587

Yahoo Mail

Host: `smtp.mail.yahoo.com` Port: 587 (requires App Password)

Office 365

Host: `smtp.office365.com` Port: 587

Custom domain

Check your hosting provider's SMTP documentation

SendGrid

Host: `smtp.sendgrid.net` Port: 587 User: `apikey` Pass: your API key

Note

SMTP credentials are stored encrypted in `data/db.json` on the server. The password is never sent back to the browser after saving.

10 — Troubleshooting

■ Server won't start — "JWT_SECRET missing"

Check your `.env` file exists and contains `JWT_SECRET` with at least 32 characters.

Run: `cat .env`

■ Server won't start — "Cannot find module"

Dependencies not installed. Run: `npm install`

■ "Failed to fetch" in the browser

You opened `index.html` as a file (`file://`) instead of through the server.

Open `http://localhost:3456` or your domain.

■ Port 3456 already in use

Kill the old process:

`lsof -ti:3456 | xargs kill -9`

Then restart: `pm2 restart crm`

■ 503 Bad Gateway in Nginx

Node.js is not running. Check: `pm2 status`

Start it: `pm2 start ecosystem.config.js`

■ SSL certificate error

Certificate may be expired. Renew: `sudo certbot renew`

Check auto-renewal: `sudo certbot renew --dry-run`

■ Login always says "Incorrect password"

Account may be locked (5 failed attempts). Wait 15 minutes or restart the server to reset.

■ Emails not sending

1. Check Email Settings in the app – should show ■
2. For Gmail, confirm App Password is used (not your main password)
3. Check server logs: `pm2 logs crm`

■ Data not saving / lost after restart

Check data/ folder exists and is writable:

```
ls -la data/
```

```
chmod 700 data/ && chmod 600 data/db.json
```

■ High CPU / memory usage

Check PM2: `pm2 monit`

Restart if needed: `pm2 reload crm`

Consider upgrading your VPS plan.

Viewing server logs

Live logs (PM2)

```
pm2 logs crm
```

Last 200 lines

```
pm2 logs crm --lines 200
```

Error log only

```
tail -f /home/user/crm/logs/err.log
```

Check if Node process is running

```
ps aux | grep node
```